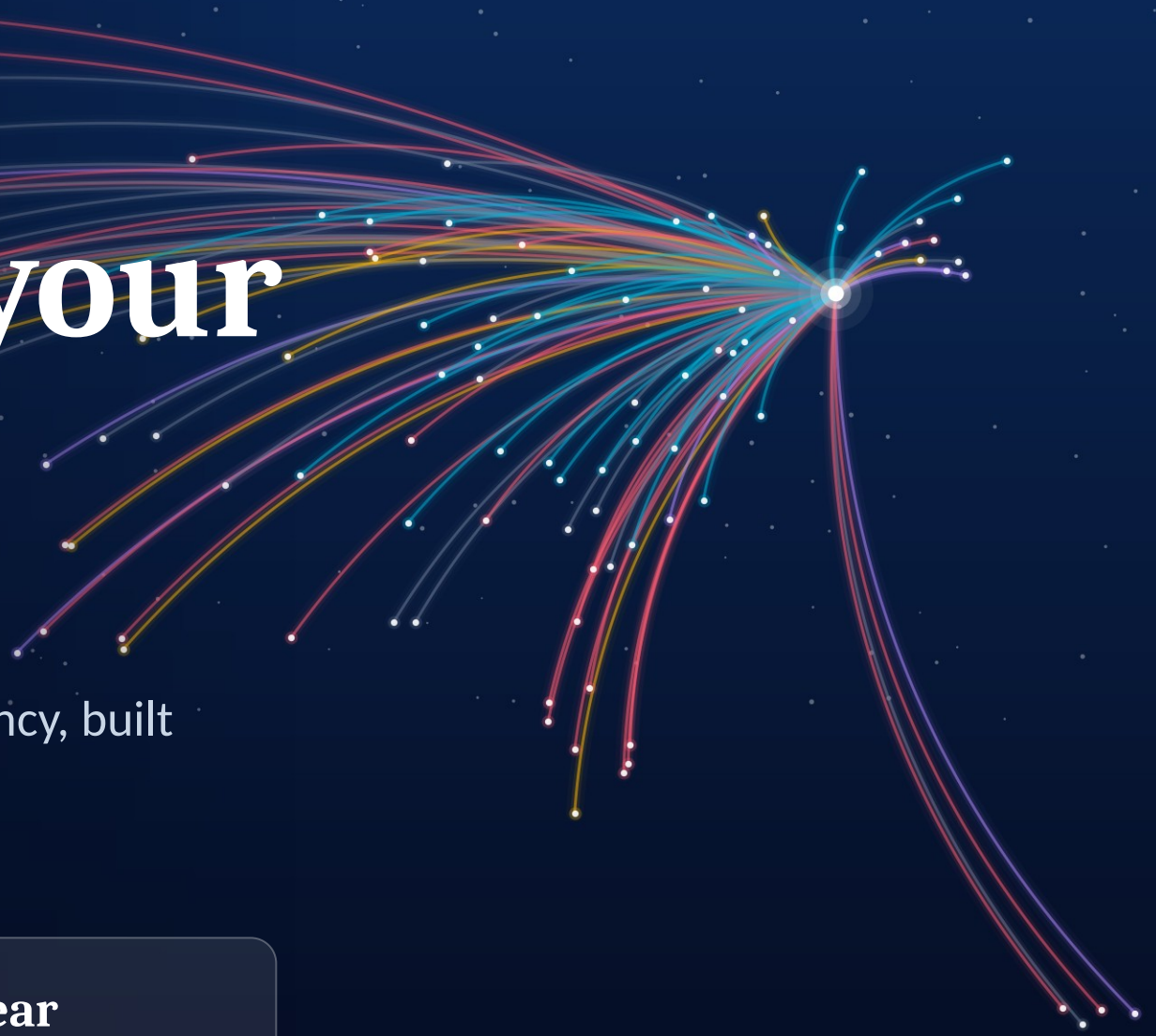
TMS · DATA & ML TEAM

Know before your clients do.



A flight early-warning system for our travel agency, built on 435,352 real departures from New York

2 hours
ahead of departure

2 alerts
late · cancelled

1 year
of real flight data

THE PROBLEM

Every late or cancelled flight is a client we can lose

23%

of flights leaving New York in 2023 were late by more than 15 minutes or cancelled

10,738 cancellations · 435,352 flights

Missed connections

One late departure can break a whole trip — and the hotel, transfer and guide behind it.

Rebooking in a hurry

A cancellation found late means expensive seats, long phone calls and a lost day.

Clients who remember

Travellers do not blame the airline alone: they remember who sold them the trip.

THE SOLUTION

An early-warning system that speaks business: what to do, and how sure we are

01

Understand

What really drives delays: hour, weather, the plane's earlier flight, what happened earlier today.

02

Group

118 destinations become 4 destination types, each with its own habits and its own advice.

03

Predict

Two alerts for every booked flight, two hours before departure.

04

Act

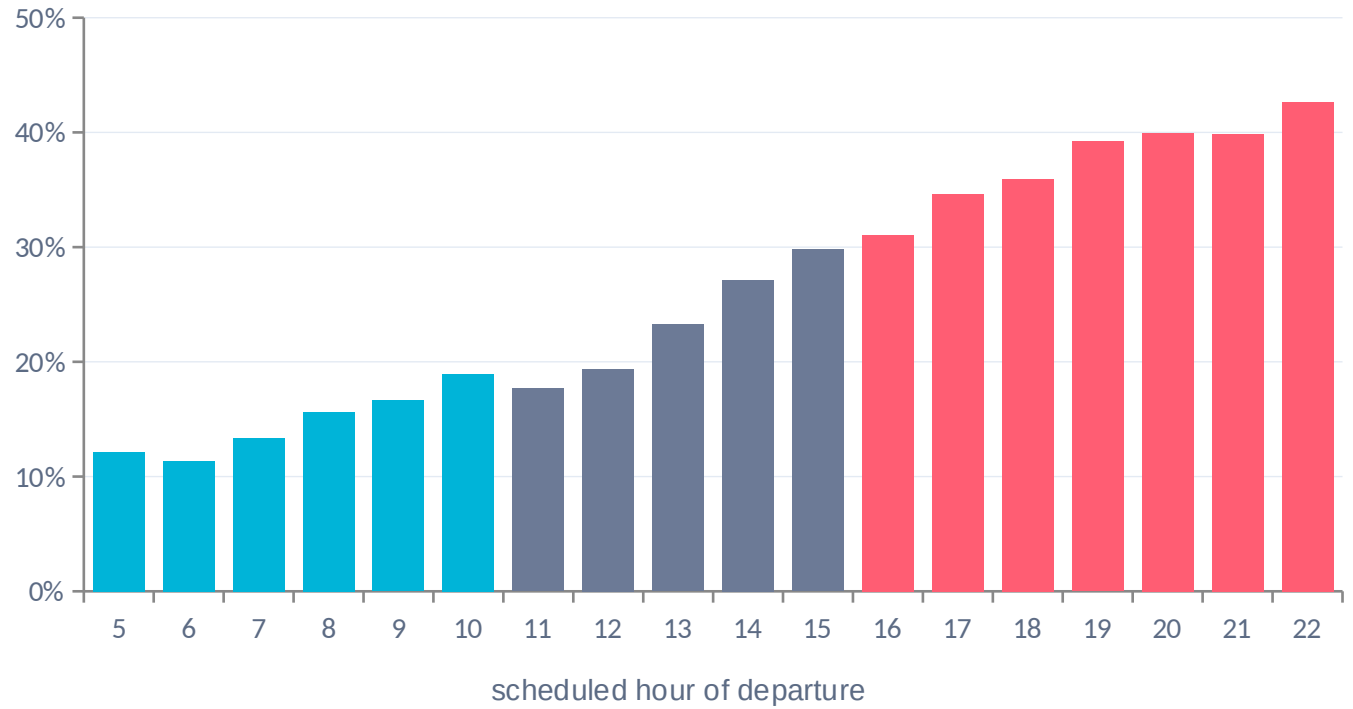
A decision card for each alert: cut-off, action, value, weak spots, when to re-check.

ALERT 1 Will this flight be more than 15 minutes late — or cancelled?

ALERT 2 Will this flight be cancelled? — the rarest, most expensive case.

WHAT THE DATA TELLS US

Mornings win, storms hurt — advice we can sell today



14% → 39%

from the first flights of the morning to the evening — and it holds inside the same month, airport and airline

Bad weather multiplies the risk

Rain or snow	**37%**	vs 24%
Visibility under 3 miles	**41%**	vs 25%
Wind gusts ≥ 25 mph	**35%**	vs 25%

Our advice: put clients with connections on morning flights.

THREE THINGS THAT LOOKED STRANGE

We checked three “common sense” ideas — and the data disagreed

“Heat delays flights”

Not really

Hot hours do look worst (43% vs 25%). But once month and hour are held equal, the effect is gone (odds ratio 2.3 → 1.0): they are summer afternoons.

Do not use temperature as a rule.

“The most punctual airline is the most reliable”

Not always

Republic is late only 13% of the time — but it cancels 4.3% of its flights, among the highest. JetBlue is the worst overall (35%), Republic the best (18%) only when we count late and cancelled together.

Judge airlines on both.

“Cancellations are random”

No — they come in waves

10 days out of 273 hold 35% of all cancellations, hitting all three airports at once. Weather at our airports explains only part of it.

What happened earlier today is the best warning.

KNOW YOUR DESTINATIONS

118 destinations, four personalities



Every line is a route from New York. Groups were learned with k-means on 82 destinations (Jan–Sep only) and still separate Jul–Sep flights by 12 points.

Short-haul regional

32 airports · fewest delays (18%), most cancellations (4.5%)

Keep an alternative ready.

Calm mornings

10 airports · mornings 12% late, evenings slip by 25 points

Offer an earlier flight.

Few flights a day

10 airports · late from the very first flight (24%)

Advise a buffer day.

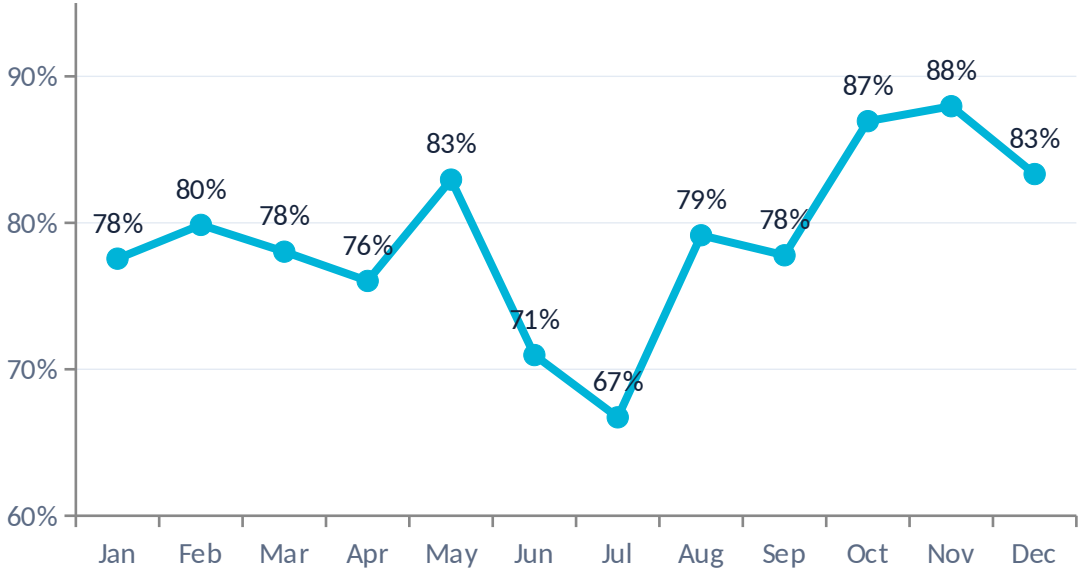
Busy hubs and leisure

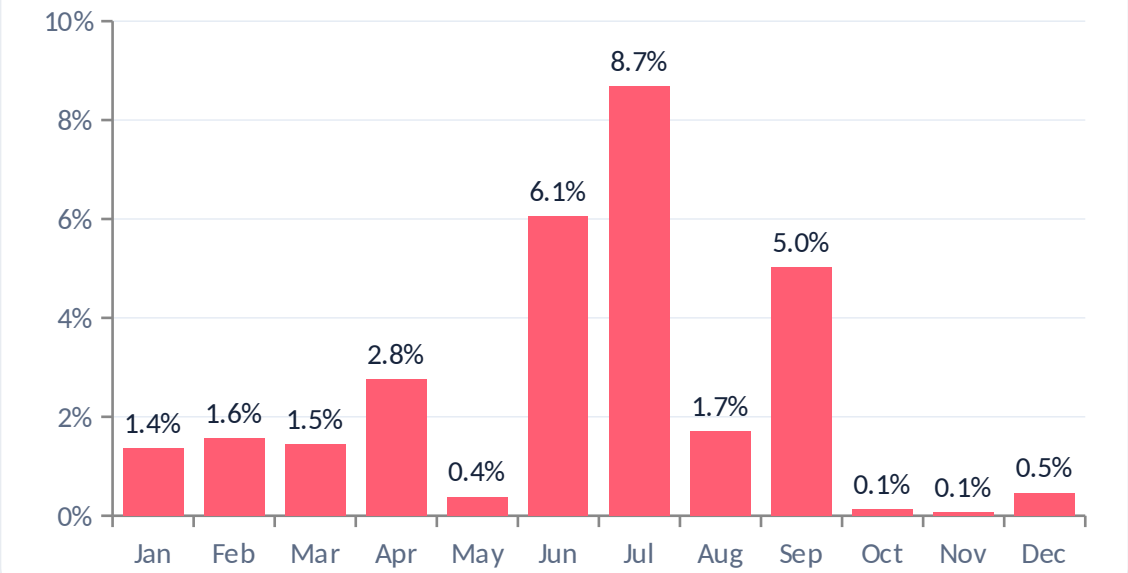
30 airports · 27% late, 59% of all flights

Book before noon.

THE DASHBOARD

Six numbers every month: now we can see the year clearly





July

worst month: 67% on time, 8.7% cancelled

Oct–Nov

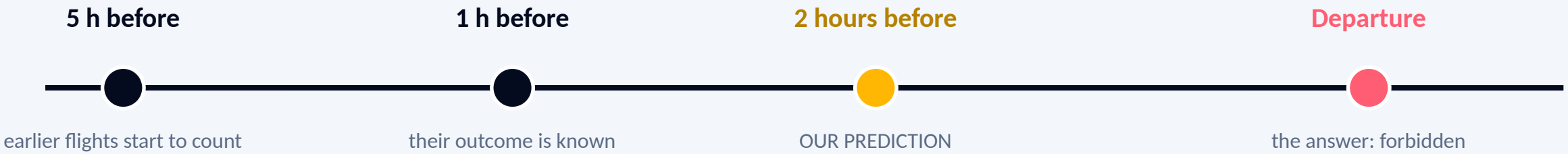
suspiciously calm: 21–48 cancellations a month

8.3 points

error of the best forecast: one year cannot teach the seasons

BUILT TO BE TRUSTED

It only uses what we really know two hours before departure



Calendar & schedule

weekday, time of day, holidays, how busy the hour is

Latest weather

visibility, wind, rain — last observation, not a forecast we do not have

The plane's earlier flight

did it leave late? is it overdue? was it cancelled?

State of the airport

how many earlier flights were late or cancelled

We hunted for cheating — and removed three leaks

- A missing tail number appears only on cancelled flights
- Filling unknown planes with a constant left a footprint (0.089 vs 0.047)
- Weather of the departure hour would need a perfect forecast

600 flights re-computed by brute force: 0 differences

A HONEST TEST

We tested it like real life: learn from the past, take the exam on the future

LEARN
January – July

CHOOSE
August – September

EXAM
October – December · once



A dumb answer to beat

“Nobody is late” is 86% accurate — and finds nothing. So we never use accuracy: we use precision, recall and PR-AUC against the dumb answer.



Nothing from the future

Every removed column is listed with how much it would give away: the real delay alone predicts the target perfectly (AUC 1.0).



Three kinds of model

Logistic regression, gradient boosting and a neural network — three different ideas, one fair comparison.

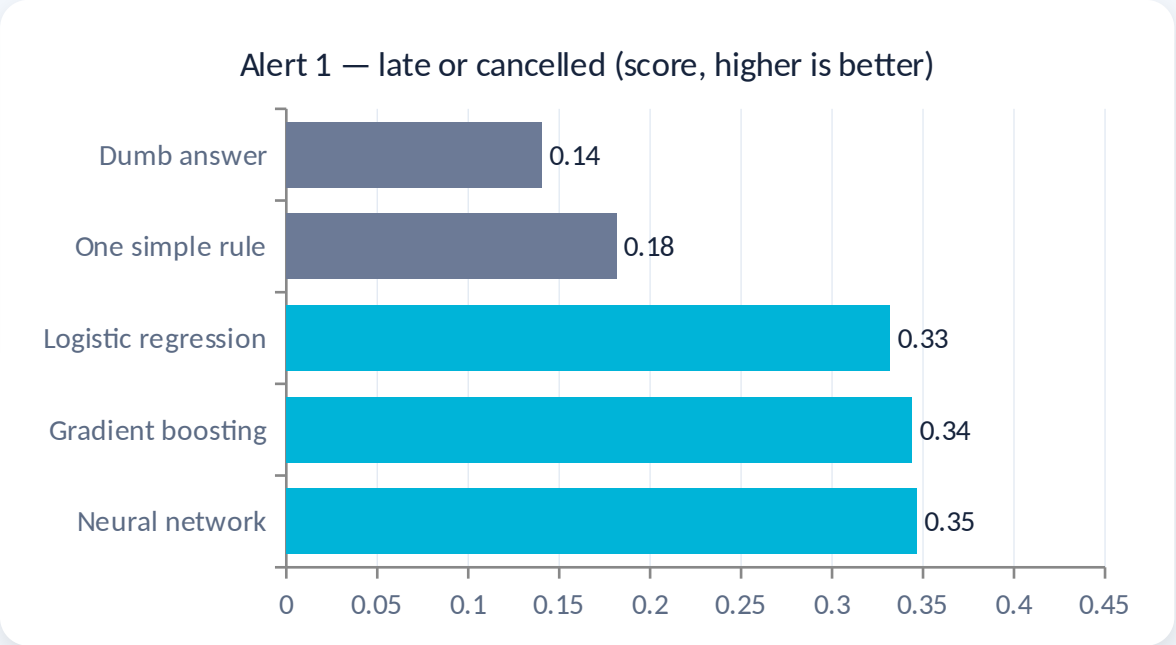


Honest about uncertainty

Intervals resample whole days; settings and cut-offs are chosen on August–September, never on the exam.

THE RESULT

On the exam months, the alerts are 3× more precise than guessing





3.2×

more precise than flagging flights at random (44% vs 14% right)

24%

of all 14,685 late or cancelled flights reached by flagging just 7.6% of the flights

72×

more precise for cancellations (15% vs 0.2%) — with only 224 events, so read with care

DECISION CARD 1

Alert 1 — flag a flight when its risk score passes 0.37

1 in 13

flights flagged on the exam months (planned: 1 in 5 — a calm autumn)

44%

of the flagged flights really late or cancelled (random: 14%)

24%

of all bad flights reached (random: 7.6%)

Action	Message the client, check the connection, keep an alternative ready. Group 3: advise a buffer day.
Weak spots	Morning flights almost never flagged (recall 3%). Too pessimistic for United (+8.7 points) and Newark (+6).
Re-check every week	if the score's lift falls under 1.5× the dumb answer (it was 2.1–2.8×), or predicted-vs-real risk differs by more than 5 points two weeks running.
Human check	A person confirms the airline's live status before a client changes plans. No alert on a morning flight does not mean “safe”.
Finding	When the plane's earlier flight left over an hour late, 71% of the next flights are late or cancelled — against 12% when it left on time.

DECISION CARD 2

Alert 2 — hold an alternative when the cancellation score passes 0.11

571

flights flagged on the exam months (0.55%, 1 in 183)

15%

of the flagged flights really cancelled (random: 0.2%)

39%

of the 224 cancellations found (88) — wide uncertainty

Action

Hold an alternative seat or start a rebooking and tell the client early. Group 1: hold a later flight of the day.

Weak spots

Finds December (57% of 155) but none of October's and November's 69. Weak at Newark: 8% vs 45–47% at JFK and LGA.

Re-check every week

on a 4-week window: if it finds under 20% of at least 10 cancellations (true in 7 of 10 windows). Until then, use it as a storm-day warning.

Human check

Check the airline's cancellation announcements and airport status, and whether the ticket can be changed, before holding a seat.

Finding

When 10% or more of the airport's earlier flights were cancelled, 10% of the next flights are cancelled — against 0.17% otherwise.

WHAT WE NEED FROM YOU

From prototype to pilot in 90 days

Weeks 1–4

Pilot

Run both alerts on a sample of real bookings. The operations team follows up; we count rebookings and complaints avoided.

Weeks 3–8

Better data

Add 2+ years of flights, a weather forecast feed, and our real costs of a rebooking and a missed connection.

Weeks 8–12

Scale

Retrain every quarter, run the weekly health check, connect the alerts to the booking system.

The capacity and cost figures on the decision cards are our assumptions: the pilot replaces them with yours.

Three decisions we ask for

1. Approve a 4-week pilot with the operations team
2. Share real costs: rebooking, missed connection, team capacity
3. Name an owner for the weekly health check

BEING STRAIGHT WITH YOU

What this is — and what it is not yet

What it is

- A tested early-warning prototype: 2.4× better than guessing for late or cancelled flights
- Alerts with a clear action, a cut-off and a weekly health check
- Fully reproducible: every number is in the repository

What it is not (yet)

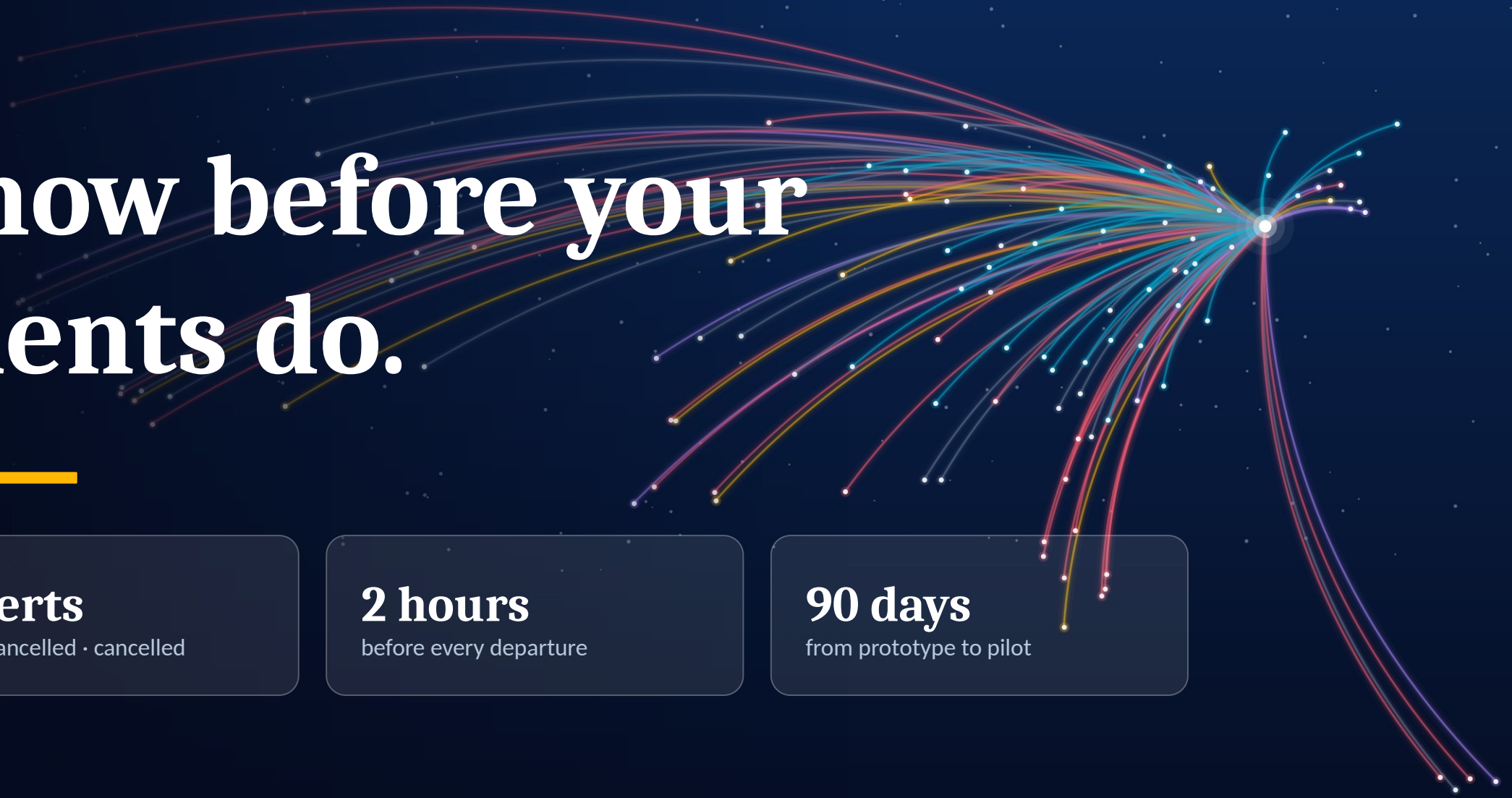
- Proven on one year of New York departures only
- Calm exam months: only 224 cancellations, so alert 2 is a storm-day warning
- Blind to storms on the route, and to morning delays

How we used AI

Claude drafted the code and text. We re-ran every notebook from the raw files and checked each number against the output — which caught real errors, including a leak hidden in how unknown planes were filled.

Our recommendation: start the pilot with alert 1 (late or cancelled), which is solid, and use alert 2 (cancelled) as a storm-day warning while we collect more data.

Know before your clients do.



2 alerts
late or cancelled · cancelled

2 hours
before every departure

90 days
from prototype to pilot

Questions?

Every number in this deck is reproducible from the project repository.